Munjung Kim

munjungkim.github.io

Interest

Computational social science, Science of science, Graph learning, Brain Network

I am broadly interested in how people learn and share information, construct their knowledge networks and collaborate to make scientific progress. I hope to explore these questions through the application of natural language processing, network science and machine learning.

Education

Pohang University of Science and Technology, Pohang, S.Korea 2017– 2022 (Expected)
Bachelor of Science, Physics

- *Relevant Coursework:* App. of Mathematics and Big Data (MATH532), Probability & Statistics(MATH230), AI for Bio-Health (LIFE451Y), Social Issue & Data Analysis(CMCC399), Statistical Mechanics(PHYS512), R Programming for Data Analysis (IMEN491R)
- GPA : 3.60/4.00 (Total), 3.83/4.00 (Last four semesters)

Oral Presentation

M.J. Kim, J. Yoon, H. Kim and W.-S. Jung, "Neural embedding of research papers reveals characteristics of policy research institutes" Networks 2021: A Joint Sunbelt & NetSci Conference,2021. [abstract][video][slide]
M.J. Kim, J. Yoon and W.-S. Jung, "Quantitative Analysis on Institutions' Activity Policy Research with Embedding Method", Korea Academy of Complexity Studies, 2019, Yonsei University, S. Korea, Nov 23, 2019.

Working Paper

M.J. Kim, J. Yoon, H. Kim and W.-S. Jung, "Quantifying the topic disparity of scientific articles." [draft]

Research Expreience

Research Internship Pohang University of Science and Technology
May 2019 – Aug 2021 *Pohang, S. Korea*

- Advisor: Jisung Yoon, Prof. Hyunuk Kim, Prof. Woo-Sung Jung
- Research Topic: Quantitative analysis on institutions' activity by using embedding method (Senior Thesis; 2nd prize in Senior Thesis Competition)

Global Talent Attraction Program Indiana University
Apr 2020 – *Virtual*

- Advisor: Youngheun Jo, Prof. Yong-yeol Ahn
- Research Topic: Finding correlation between edge-Functional Connectivity and behavior data
- Compared behavioral explainability between eFC and FC using heritability and CPM (Connectome-based predictive modeling)

Research and Education Program for High School Students Pohang University of Science and Technology
Pohang, S. Korea
Apr 2015 – Aug 2015

- Advisor: Prof. Joung-Hun Kim
- Research Topic: A Comparative study on fear conditioning depending on 3 sense using c-Fos

Honors &Awards

The National Scholarship for Science and Engineering, 2017-2021, S. Korea Government
Global Talent Attraction Program, Indiana University, 2020 (Postponed due to Covid)
Senior Thesis Award, 2nd Prize
Semester of High Honor, 2019,2020,2021

TA Expreience

PHYS312: Computational Physics, 2019
MATH 102: Calculus 2, 2019, 2020

Work Expreience

SK Hynix, Intern, Quality Control Department, Feb 2019

Skill

Languages: English (Proficient/ TOEFL 105), Korean(Native), Japanese(Elementary)
Computer Software: Python (Main), R, Matlab, LaTeX

Extracurricular Activity

Residential Advisor

- Worked as mentor of students and conducted conflict resolution hearings between students resulting in a professional atmosphere
- Planned and conducted residential activities [Link]
- Managed residential regulations reflecting students' opinion

POSTECH Cheerleading Club

- Led cheers in 2017 Postech-Kaist annual sports competition [Link]

POSTECH Readers Club

- Worked as a mentor of the students in the reading club and organized book discussion forums
- Discussed books : Story of Your Life (Ted Chiang), 선량한 차별주의자 (Ji-hye Kim), If we can't go at the speed of light(Choyeop Kim), Breath (Ted Chiang)

International Quant Championship

- Participated a team based alpha-building competition hosted by WorldQuant.
- Ranked 10 out of 451 teams in South Korea

References

Dr. Woo-Sung Jung, *Associate Professor*
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wsjung@postech.ac.kr

Dr. Yong-Yeol Ahn, *Associate Professor*
Luddy Shool of Informatics, Computing, and Engineering
Indiana Unversity Bloomington, Bloomington, IN 47409, USA
yyahn@iu.edu

Dr. Hyunuk Kim, *Assistant Professor*
Administrative Sciences, Metropolitan College
Boston University, Boston, MA 02215, USA
uk@bu.edu